FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no.23

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Recap *of previous lecture*

PERCENTAGE



Topics to be covered

"AVERAGE & WEIGHTED AVERAGE"



① Average $(\bar{x}) = \dfrac{\text{Sum of observations}}{\text{Number of observations}} = \dfrac{x_1+x_2+x_3+\ldots\ldots+x_n}{n}$

ie $\boxed{\bar{x} = \dfrac{\Sigma x}{N}}$

$\Rightarrow \Sigma x = N(\bar{x})$

Note: Average of an AP $= \left(\dfrac{a+l}{2}\right)$

② weighted Average → Let C_1 & C_2 are the Quantities whose Average is being found & w_1 & w_2 are their weights then

$$\text{Weighted Average} = \frac{C_1 \times w_1 + C_2 \times w_2}{w_1 + w_2}$$

Q: Find the Average of all the terms in the series ?

$1-2+3-4+5-6+7-8+\cdots\cdots+197-198+199-200$

Sol: In this series $n = 200$ terms

$X = \{1, -2, 3, -4, 5, -6, \cdots\cdots, 199, -200\}$

$\Sigma X = 1+(-2)+3+(-4)+5+(-6)+\cdots\cdots+199+(-200)$

$= (1-2)+(3-4)+(5-6)+\cdots\cdots+(197-198)+(199-200)$

$= -1+(-1)+(-1)+(-1)+\cdots\cdots+(-1)+(-1)$ (100 times)

$= -100$

$\therefore \bar{X} = \frac{\Sigma X}{n} = \frac{-100}{200} = -0.5$

Q: In an examination, Average of 40 students is 72. Afterwards it is found that marks of 3 students are misread as 68, 65 & 73 instead of 64, 62 & 84 resp. find the Correct average?

Sol: x = Marks obtained

$n = 40$ students.

Wrong Sum $(\Sigma x) = n\bar{x} = 40 \times 72 = 2880$

Correct Average $= \dfrac{(\text{wrong sum}) - (68+65+73) + (64+62+84)}{n}$

(Wrong Av = 72)

$= \dfrac{2880 - 206 + 210}{40} = 72.1$

Q3. Three years ago, Average age of family of 5 members was 35 yrs. One year ago, a new baby was born. Three years hence, the Average age of the family will be?

Ex = Av × n

Sol: three year ago, Total age of family of 5 members = $35 \times 5 = 175$ yrs

one year ago, " " " " 6 " $= 175 + 2\times5 + 0 = 185$ yrs.

Currently, " " " " 6 " $= 185 + 6 = 191$ yrs

Three yrs Hence, Total age " " 6 " $= 191 + 3\times6 = 209$ yrs.

$$\therefore \text{Av. age} = \frac{\text{Total age}}{\text{No of Members}} = \frac{209}{6} = 34\frac{5}{6} \text{ yrs.}$$

eg: Amit borrowed Rs 6000 at 4% rate and Rs 4000 at 6% rate.
If the Interest rate are Simple Interest then find the Av. ROI, Amit has to pay on the entirely borrowed amount at the end of year

(M-I) Interest paid by Amit on Rs 6000 = ? $= 6000 \times \frac{4}{100} = 240$ Rs

" " " " Rs 4000 = ? $= 4000 \times \frac{6}{100} = 240$ Rs.

Total Interest paid by Amit on entirely borrowed amount = 480 Rs

(6000 + 4000 = 10000)

∵ 480 is how much % of 10000

$\Rightarrow 480 = \frac{?}{100} \times 10000 \Rightarrow ? = 4.8\%$

∴ Av ROI = 4.8%

Method II Wt Average $= \frac{C_1W_1 + C_2W_2}{W_1 + W_2} \Rightarrow$ AN ROI $= \frac{4\% \times 6000 + 6\% \times 4000}{6000 + 4000}$

$C_1 = 4\%$, $W_1 = 6000$

$C_2 = 6\%$, $W_2 = 4000$

$= \frac{24\% + 24\%}{10} = \frac{48\%}{10} = 4.8\%$

Q. 9 ltrs of Milk & water sol (having 40% Milk) is mixed with 3 ltrs of Milk & water sol (having 70% Milk). then find the % of Milk in the resulting Mixture?

Sol:

$C_1 = 40\%$, $W_1 = 9$ ltrs

$C_2 = 70\%$, $W_2 = 3$ ltrs

So Final % of Milk $= \dfrac{C_1W_1 + C_2W_2}{W_1 + W_2}$

$= \dfrac{40\% \times 9 + 70\% \times 3}{9+3}$

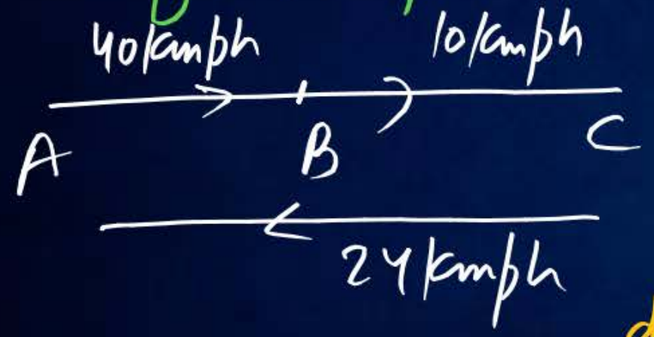
$= \dfrac{360\% + 210\%}{12} = \dfrac{570\%}{12}$

$= 47.5\%$

Q. A person went from A to B at a speed of 40 kmph & B to C at a speed of 10 kmph (where B is mid point of A & C). The he returned from C to A (via B) at a speed of 24 kmph then find the Average speed for the entire journey?

$S = \frac{D}{T}$ or $D = S \times T$

$t = \frac{D}{S}$



M-I

$d_1 : d_2 : d_3 = 1 : 1 : 2$

$d_1 = k$

$d_2 = k$

$d_3 = 2k$

$$\text{Av. Speed} = \frac{\text{Total Distance Covered}}{\text{Total time taken}} = ?$$

$$= \frac{d_1 + d_2 + d_3}{t_1 + t_2 + t_3} = \frac{d_1 + d_2 + d_3}{\left(\frac{d_1}{s_1} + \frac{d_2}{s_2} + \frac{d_3}{s_3}\right)}$$

$= \text{-----} = 19.2 \text{ kmph}$

M-II

Av speed for Entire journey $= \frac{\text{Total Dist}}{\text{Total time}} = \frac{d_{AB}+d_{BC}+d_{CA}}{t_{AB}+t_{BC}+t_{CA}}$

→ 40 kmph ← 10 kmph



← 24 kmph

$$\text{Av speed} = \frac{S_1 \times t_1 + S_2 \times t_2 + S_3 \times t_3}{t_1 + t_2 + t_3}$$

$$= \frac{40(3K) + 10(12K) + 24(5K)}{(3K) + (12K) + (5K)}$$

$= 19.2$ kmph

$S = \frac{D}{t}$ & $\boxed{D = S \times t}$ & $t = \frac{D}{S}$

$d_1 : d_2 : d_3 = 1 : 1 : 2$

$S_1 : S_2 : S_3 = 40 : 10 : 24$

$\Rightarrow t_1 : t_2 : t_3 = \frac{1}{40} : \frac{1}{10} : \frac{1}{24} = \frac{120}{40} : \frac{120}{10} : \frac{120}{24} = 3 : 12 : 5$

Q. Average score obtained by students in three classes X, Y & Z are 83, 76 & 85 resp. The Average of score of all the students in Class X & Y together is 79 and Y & Z Together is 81. what is the Average score of all 3 classes?

Sol: $Av = \frac{\Sigma x}{n} \Rightarrow \boxed{\Sigma x = Av \times n}$, Now A.T.Q.

$\Sigma X = (83)x$, $\Sigma Y = (76)y$, $\Sigma Z = (85)z$

again $\Sigma(X+Y) = 79(x+y)$ & $\Sigma(Y+Z) = 81(y+z)$

$\Sigma X + \Sigma Y = 79x + 79y$ | $\Sigma Y + \Sigma Z = 81y + 81z$

$83x + 76y = 79x + 79y$ | $76y + 85z = 81y + 81z$

$4x = 3y$ | $4z = 5y$

$4x = 3y$ & $4z = 5y$

$\frac{x}{y} = \frac{3}{4}$ & $\frac{y}{z} = \frac{4}{5}$

$x : y = 3 : 4$ & $y : z = 4 : 5$

$\Rightarrow \boxed{x : y : z = 3 : 4 : 5}$

$\Rightarrow x = 3k,\ y = 4k,\ z = 5k$

i.e. $w_1 = 3k,\ w_2 = 4k,\ w_3 = 5k$

& $C_1 = 83,\ C_2 = 76,\ C_3 = 85$

Av score of all three classes

$$= \frac{C_1 w_1 + C_2 w_2 + C_3 w_3}{w_1 + w_2 + w_2}$$

$$= \frac{83(3k) + 76(4k) + 85(5k)}{(3k) + (4k) + (5k)}$$

$= 81.5$ Marks.


Thank
YOU